has the form of an inner product. If the column vector α is orthogonal to ψ, then the left side of Eq. 7.26 is zero. Such a vector is an eigenvector of ρ with eigenvalue zero.

If the dimension of Alice's space of states is N_A, then there are $N_A - 1$ vectors orthogonal to ψ. Each one of them is an eigenvector of ρ with eigenvalue 0. That leaves only one possible direction for an eigenvector with a nonzero eigenvalue, namely the vector $\psi(a)$. In fact, if we plug in $\alpha_a = \psi(a)$, we do indeed find that it is an eigenvector of ρ with eigenvalue 1.

To summarize the theorem: If the composite Alice-Bob system is in a product state, then Alice's (or Bob's) density matrix has one and only one eigenvalue equal to 1, and all the rest are zero. Moreover, the eigenvector with a nonzero eigenvalue is nothing but the wave function of Alice's half of the system.

In this situation, Alice's system is in a pure state. All of Alice's observations are described as if Bob and his system never existed and Alice had an isolated system described by the wave function $\psi(a')$.

The opposite extreme of a pure state is a *maximally entangled* state. Maximally entangled states are states of a combined system in which nothing is known about either subsystem, even though they are complete descriptions of the system as a whole—as complete as quantum mechanics allows. The state $|sing\rangle$ is a maximally entangled state.

When Alice calculates her density matrix for a maximally entangled state, she finds something very disappointing: the density matrix is proportional to the unit matrix. All the eigenvalues are equal, and given that they all sum to unity, each eigenvalue is equal to $1/N_A$. In other words,

$$\rho_{a'a} = \frac{1}{N_A}\delta_{a'a}. \qquad (7.27)$$

Why is Alice disappointed? Go back to Eq. 7.22. This equation says that the probability for a particular state a is the diagonal element of ρ, but Eq. 7.27 tells us that all the probabilities are equal. What could be less informative